Hundred Days of Coding in C programing

Day-14

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Batch - 15

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Prepare the algorithm and test case table.

Then write, compile, execute, and test the C program against the prepared use cases.

Write a menu-driven C program using switch case to input a positive integer and perform one of the following operations based on the user's choice:

Oquestion 1

Check Whether All Digits Are Unique

Step 1 - Algorithm

1. Start.
2. Input a positive integer n.
3. Initialize freq[10] = 0.
4. Store n in a temporary variable temp.
5. Extract the last digit using digit = temp % 10.
6. Increase the frequency of that digit: freq[digit]++.
7. Remove the last digit using temp = temp / 10.
8. Repeat Steps 5 to 7 until temp becomes 0.
9. Check the frequency of every digit from 0 to 9.
10. If any digit has a frequency greater than 1, set unique = 0.
11. If unique = 1, display "Unique"; otherwise, display "Not Unique".
12. Stop.

Step-2 Test Case Table

Test Case	Input Number	Expected Output	Actual Output
TC1	12345	Unique	Unique
TC2	12234	Not Unique	Not Unique
TC3	987654	Unique	Unique
TC4	112233	Not Unique	Not Unique

Step- 3 Write the C program

```c
#include <stdio.h>

int main()
{
   int n, choice, temp, digit;
   int freq[10] = {0};
   int unique = 1;

   printf("Enter a positive integer: ");
   scanf("%d", &n);

   printf("\nMENU\n");
   printf("1. Check whether all digits are unique\n");
   printf("Enter your choice: ");
   scanf("%d", &choice);

   switch (choice)
   {
      case 1:
         temp = n;

         while (temp > 0)
         {
            digit = temp % 10;
            freq[digit]++;
            temp = temp / 10;
         }

         for (int i = 0; i < 10; i++)
         {
            if (freq[i] > 1)
            {
               unique = 0;
               break;
            }
```

```
        }

        if (unique)
            printf("Unique\n");
        else
            printf("Not Unique\n");
        break;

    default:
        printf("Invalid choice\n");
    }

    return 0;
}
```

Step 4- Compile and Execute the Program

Case :1

```c
{
    int n, choice, temp, digit;
    int freq[10] = {0};
    int unique = 1;
    printf("Enter a positive integer: ");
    scanf("%d", &n);
    printf("\nMENU\n");
    printf("1. Check whether all digits are unique\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);
    switch (choice)
    {
        case 1:
            temp = n;
            while (temp > 0)
            {
                digit = temp % 10;
                freq[digit]++;
                temp = temp / 10;
            }
            for (int i = 0; i < 10; i++)
            {
                if (freq[i] > 1)
                {
                    unique = 0;
                    break;
                }
            }
            if (unique)
                printf("Unique\n");
            else
                printf("Not Unique\n");
            break;
        default:
            printf("Invalid choice\n");
    }
    return 0;
}
```

```
Enter a positive integer: 12345

MENU
1. Check whether all digits are unique
Enter your choice: 1
Unique
```

Case :2

```c
{
    int n, choice, temp, digit;
    int freq[10] = {0};
    int unique = 1;
    printf("Enter a positive integer: ");
    scanf("%d", &n);
    printf("\nMENU\n");
    printf("1. Check whether all digits are unique\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);
    switch (choice)
    {
        case 1:
            temp = n;
            while (temp > 0)
            {
                digit = temp % 10;
                freq[digit]++;
                temp = temp / 10;
            }
            for (int i = 0; i < 10; i++)
            {
                if (freq[i] > 1)
                {
                    unique = 0;
                    break;
                }
            }
            if (unique)
                printf("Unique\n");
            else
                printf("Not Unique\n");
            break;
        default:
            printf("Invalid choice\n");
    }
    return 0;
}
```

```
Enter a positive integer: 12234

MENU
1. Check whether all digits are unique
Enter your choice: 1
Not Unique
```

Case :3

```c
{
    int n, choice, temp, digit;
    int freq[10] = {0};
    int unique = 1;
    printf("Enter a positive integer: ");
    scanf("%d", &n);
    printf("\nMENU\n");
    printf("1. Check whether all digits are unique\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);
    switch (choice)
    {
        case 1:
            temp = n;
            while (temp > 0)
            {
                digit = temp % 10;
                freq[digit]++;
                temp = temp / 10;
            }
            for (int i = 0; i < 10; i++)
            {
                if (freq[i] > 1)
                {
                    unique = 0;
                    break;
                }
            }
            if (unique)
                printf("Unique\n");
            else
                printf("Not Unique\n");
            break;
        default:
            printf("Invalid choice\n");
    }
    return 0;
}
```

```
Enter a positive integer: 987654

MENU
1. Check whether all digits are unique
Enter your choice: 1
Unique
```

Case :4

```c
    int n, choice, temp, digit;
    int freq[10] = {0};
    int unique = 1;
    printf("Enter a positive integer: ");
    scanf("%d", &n);
    printf("\nMENU\n");
    printf("1. Check whether all digits are unique\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);
    switch (choice)
    {
        case 1:
            temp = n;
            while (temp > 0)
            {
                digit = temp % 10;
                freq[digit]++;
                temp = temp / 10;
            }
            for (int i = 0; i < 10; i++)
            {
                if (freq[i] > 1)
                {
                    unique = 0;
                    break;
                }
            }
            if (unique)
                printf("Unique\n");
            else
                printf("Not Unique\n");
            break;
        default:
            printf("Invalid choice\n");
    }
    return 0;
}
```

```
Enter a positive integer: 112233

MENU
1. Check whether all digits are unique
Enter your choice: 1
Not Unique
```

Step – 5 Conclusion

The C program was successfully compiled and executed. It correctly checks whether all digits of a given positive integer are unique. For numbers such as 12345, it displays "Unique", while for numbers such as 12234, it displays "Not Unique". All four prepared test cases produced the expected results.

Question 2

Find the Frequency of Each Digit from 0 to 9

Step – 1 Algorithm

1. Start.
2. Input a positive integer n.
3. Initialize an array freq[10] with all elements set to 0.
4. Store the value of n in a temporary variable temp.
5. Extract the last digit using digit = temp % 10.
6. Increase the frequency of that digit using freq[digit]++.
7. Remove the last digit using temp = temp / 10.
8. Repeat Steps 5 to 7 until temp becomes 0.
9. Display the frequency of each digit from 0 to 9.
10. Stop.

Step – 2 Test Case Table

Test Case	Input Number	Expected Output	Actual Output
TC1	112333	freq_0=0, freq_1=2, freq_2=1, freq_3=3, freq_4=0, freq_5=0, freq_6=0, freq_7=0, freq_8=0, freq_9=0	Same as expected
TC2	12345	freq_0=0, freq_1=1, freq_2=1, freq_3=1, freq_4=1, freq_5=1, freq_6=0, freq_7=0, freq_8=0, freq_9=0	Same as expected
TC3	10001	freq_0=3, freq_1=2, freq_2=0, freq_3=0, freq_4=0, freq_5=0, freq_6=0, freq_7=0, freq_8=0, freq_9=0	Same as expected
TC4	99999	freq_0=0, freq_1=0, freq_2=0, freq_3=0, freq_4=0, freq_5=0, freq_6=0, freq_7=0, freq_8=0, freq_9=5	Same as expected

Step – 3 Write the C Program

```
#include <stdio.h>

int main()
{
int n, choice, temp, digit;
int freq[10] = {0};

printf("Enter a positive integer: ");
scanf("%d", &n);

printf("\nMENU\n");
printf("1. Find the frequency of each digit from 0 to 9\n");
printf("Enter your choice: ");
scanf("%d", &choice);

switch (choice)
{
case 1:
temp = n;

while (temp > 0)
{
digit = temp % 10;
freq[digit]++;
temp = temp / 10;
}

printf("\nFrequency of each digit:\n");

for (int i = 0; i < 10; i++)
{
printf("freq_%d = %d\n", i, freq[i]);
}
break;

default:
printf("Invalid choice\n");
}
```

return 0;
}

Step – 4 Compile and Execute

Case: 1

```c
    scanf("%d", &n);
    printf("\nMENU\n");
    printf("1. Find the frequency of each digit from 0 to 9\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);
    switch (choice)
    {
        case 1:
            temp = n;
            while (temp > 0)
            {
                digit = temp % 10;
                freq[digit]++;
                temp = temp / 10;
            }
            printf("\nFrequency of each digit:\n");
            for (int i = 0; i < 10; i++)
            {
                printf("freq_%d = %d\n", i, freq[i]);
            }
            break;
        default:
            printf("Invalid choice\n");
    }
    return 0;
```

```
Enter a positive integer: 112333

MENU
1. Find the frequency of each digit from 0 to 9
Enter your choice: 1

Frequency of each digit:
freq_0 = 0
freq_1 = 2
freq_2 = 1
freq_3 = 3
freq_4 = 0
freq_5 = 0
freq_6 = 0
freq_7 = 0
freq_8 = 0
freq_9 = 0
```

Case: 2

```c
    scanf("%d", &n);
    printf("\nMENU\n");
    printf("1. Find the frequency of each digit from 0 to 9\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);
    switch (choice)
    {
        case 1:
            temp = n;
            while (temp > 0)
            {
                digit = temp % 10;
                freq[digit]++;
                temp = temp / 10;
            }
            printf("\nFrequency of each digit:\n");
            for (int i = 0; i < 10; i++)
            {
                printf("freq_%d = %d\n", i, freq[i]);
            }
            break;
        default:
            printf("Invalid choice\n");
    }
    return 0;
```

```
Enter a positive integer: 12345

MENU
1. Find the frequency of each digit from 0 to 9
Enter your choice: 1

Frequency of each digit:
freq_0 = 0
freq_1 = 1
freq_2 = 1
freq_3 = 1
freq_4 = 1
freq_5 = 1
freq_6 = 0
freq_7 = 0
freq_8 = 0
freq_9 = 0
```

Case: 3

```c
    scanf("%d", &n);
    printf("\nMENU\n");
    printf("1. Find the frequency of each digit from 0 to 9\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);
    switch (choice)
    {
        case 1:
            temp = n;
            while (temp > 0)
            {
                digit = temp % 10;
                freq[digit]++;
                temp = temp / 10;
            }
            printf("\nFrequency of each digit:\n");
            for (int i = 0; i < 10; i++)
            {
                printf("freq_%d = %d\n", i, freq[i]);
            }
            break;
        default:
            printf("Invalid choice\n");
    }
    return 0;
}
```

input

```
Enter a positive integer: 10001

MENU
1. Find the frequency of each digit from 0 to 9
Enter your choice: 1

Frequency of each digit:
freq_0 = 3
freq_1 = 2
freq_2 = 0
freq_3 = 0
freq_4 = 0
freq_5 = 0
freq_6 = 0
freq_7 = 0
freq_8 = 0
freq_9 = 0
```

Case: 4

```c
    scanf("%d", &n);
    printf("\nMENU\n");
    printf("1. Find the frequency of each digit from 0 to 9\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);
    switch (choice)
    {
        case 1:
            temp = n;
            while (temp > 0)
            {
                digit = temp % 10;
                freq[digit]++;
                temp = temp / 10;
            }
            printf("\nFrequency of each digit:\n");
            for (int i = 0; i < 10; i++)
            {
                printf("freq_%d = %d\n", i, freq[i]);
            }
            break;
        default:
            printf("Invalid choice\n");
    }
    return 0;
```

```
Enter a positive integer: 99999

MENU
1. Find the frequency of each digit from 0 to 9
Enter your choice: 1

Frequency of each digit:
freq_0 = 0
freq_1 = 0
freq_2 = 0
freq_3 = 0
freq_4 = 0
freq_5 = 0
freq_6 = 0
freq_7 = 0
freq_8 = 0
freq_9 = 5
```

Step -5 Conclusion

The C program was successfully compiled and executed. It correctly calculates and displays the frequency of each digit from 0 to 9 in the given positive integer. For the input 112333, the frequencies are freq_0 = 0, freq_1 = 2, freq_2 = 1, and freq_3 = 3, while the remaining digits have frequency 0.